Developing A Diabetes Prediction Model

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W207 Summer 2023, Section 2
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Agenda

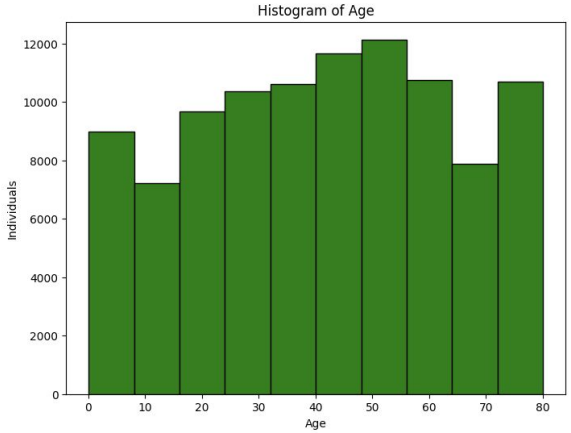
- Dataset Overview & Project Goal
- Exploratory Data Analysis
- EDA Findings
- Hypothesis
- Model Selections & Findings
- Model Tuning & Improvements
- Conclusion

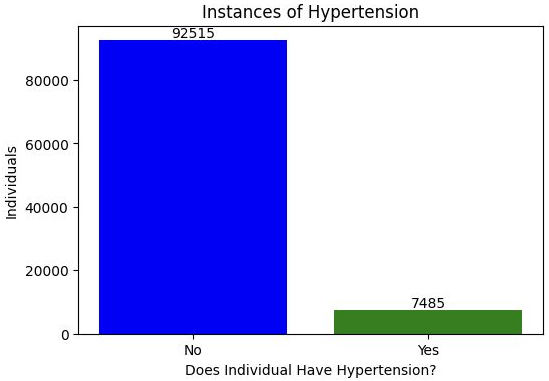
Dataset Description & Project Goal

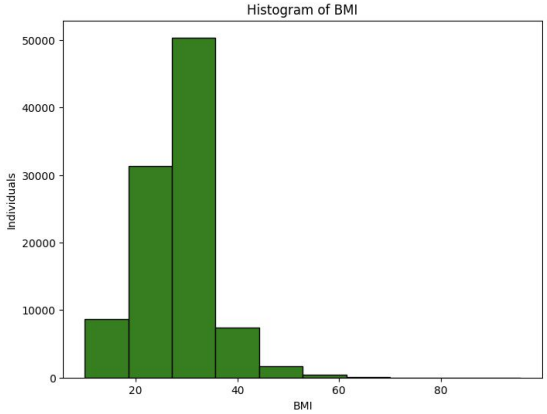
- This Kaggle dataset is an aggregation of 100K records of 8 medical and demographic fields (see below), along with a binary diabetes diagnosis field, for individuals who have been diagnosed with or at risk of developing diabetes
  - Electronic Health Records (EHRs) are the primary source of this data; EHRs were gathered from multiple healthcare providers then aggregated into one dataset
- **Goal**: Use patients' medical and demographic data to build multiple machine learning models that can predict patients' likelihood of developing diabetes
  - Understanding the relationships between key variables will enable doctors to develop personalized care plans for at-risk and diagnosed individuals

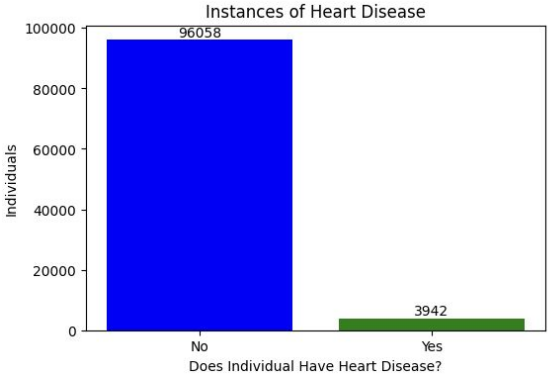
	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0
5	Female	20.0	0	0	never	27.32	6.6	85	0
6	Female	44.0	0	0	never	19.31	6.5	200	1
7	Female	79.0	0	0	No Info	23.86	5.7	85	0
8	Male	42.0	0	0	never	33.64	4.8	145	0
9	Female	32.0	0	0	never	27.32	5.0	100	0

Exploratory Data Analysis Visuals

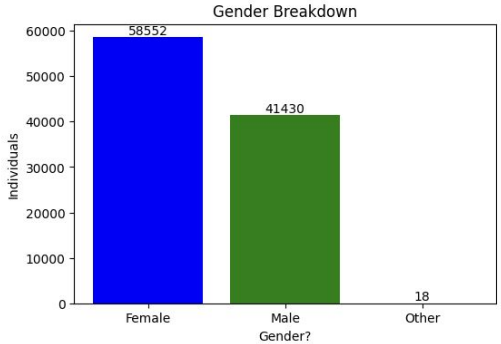


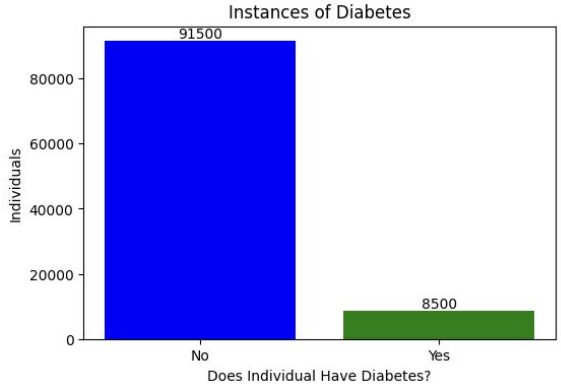


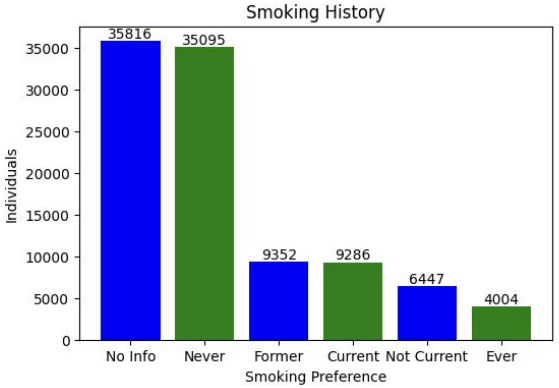


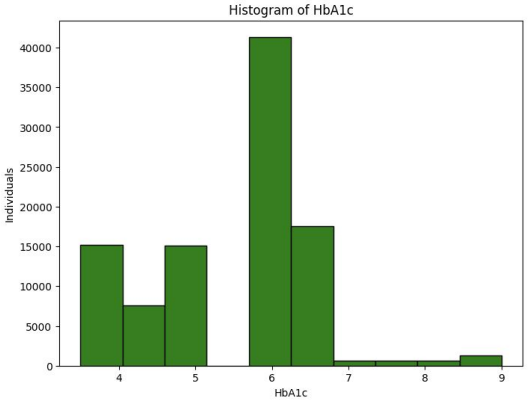


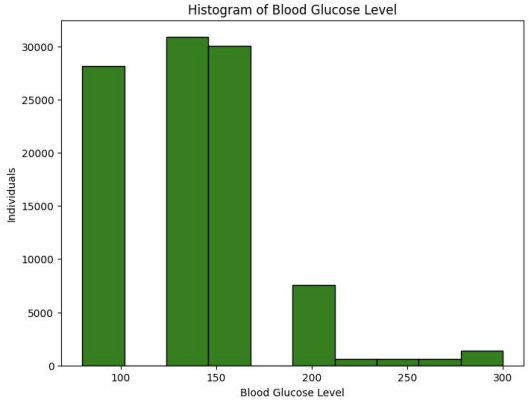
Exploratory Data Analysis Visuals Cont.











Initial Findings

- No null values
- 3.8K duplicate records → **kept**
- ~8.5% of all members in our dataset had diabetes
- Per the Mayo Clinic, individuals with an A1C between 5.7% - 6.4% means they're in the "pre-diabetes" stage
  - For modeling, we filtered our sample to members with >= 5.7% A1C
  - When filtered to "high A1C" members (>=5.7%), only 14% of those members had diabetes

Hypothesis

- Based on the Mayo Clinic, someone is considered to have diabetes if they register two A1C tests > 6.5%
- We want to train a model that looks at all the observations of people with >= 5.7% and predict whether or not they have diabetes before the second A1C test

Hypothesis: A model can be trained that will accurately determine if someone with pre-diabetes A1c level or greater has diabetes

Approach

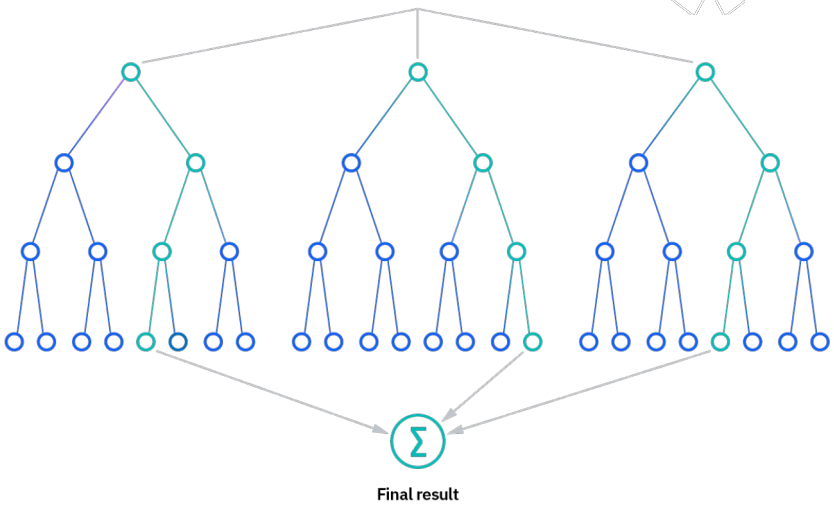
- Only focus on data where A1C >= 5.7
  - 37285
- Smoking History and Gender variables were processed using one-hot encoding
- Run sklearn's StandardScaler on the "age", "bmi", "HbA1C", "blood_glucose_level" fields

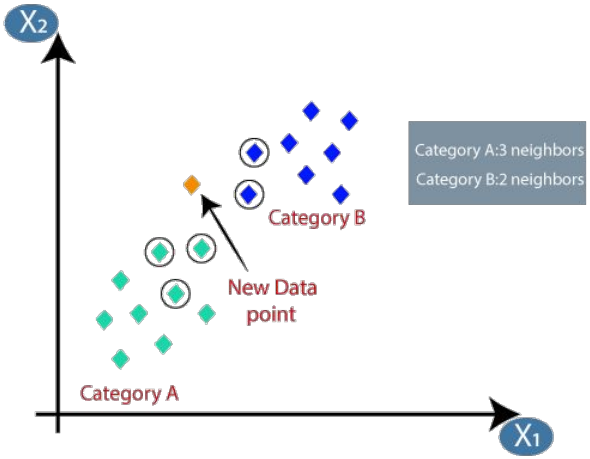
Models Used

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree Classifier

Ensembles:

- Random Forest
- Adaboost
- Gradient Boosting
- Voting Hard/Soft

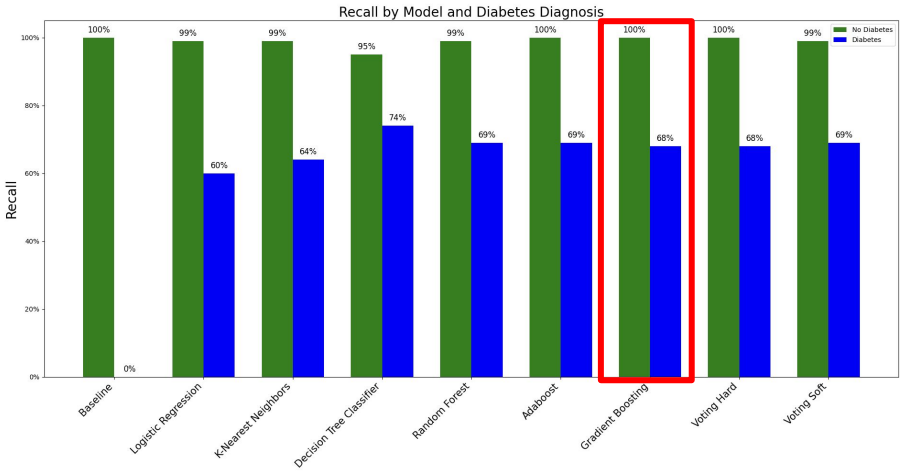


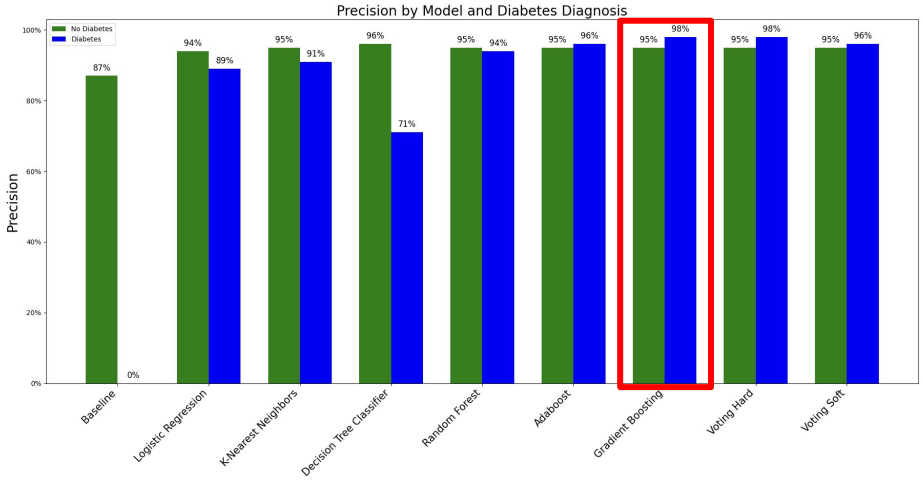


Tuning & Improvements

- Usage of ensembles improved the accuracy of predicting a diabetes diagnosis.
- Hyperparameter search did not result in a better result
- Undersampling did not help in improving the recall of predicting diabetes

Model Findings





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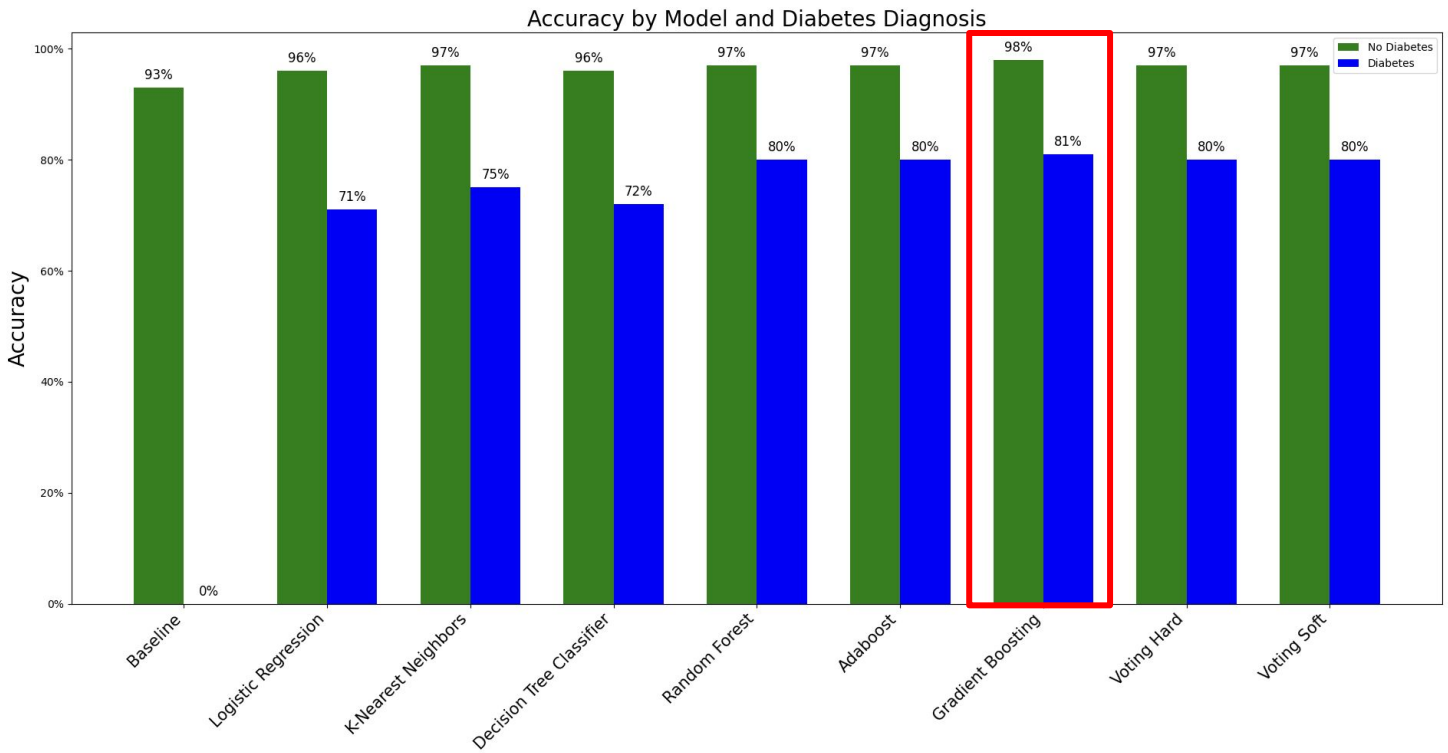
Model Findings

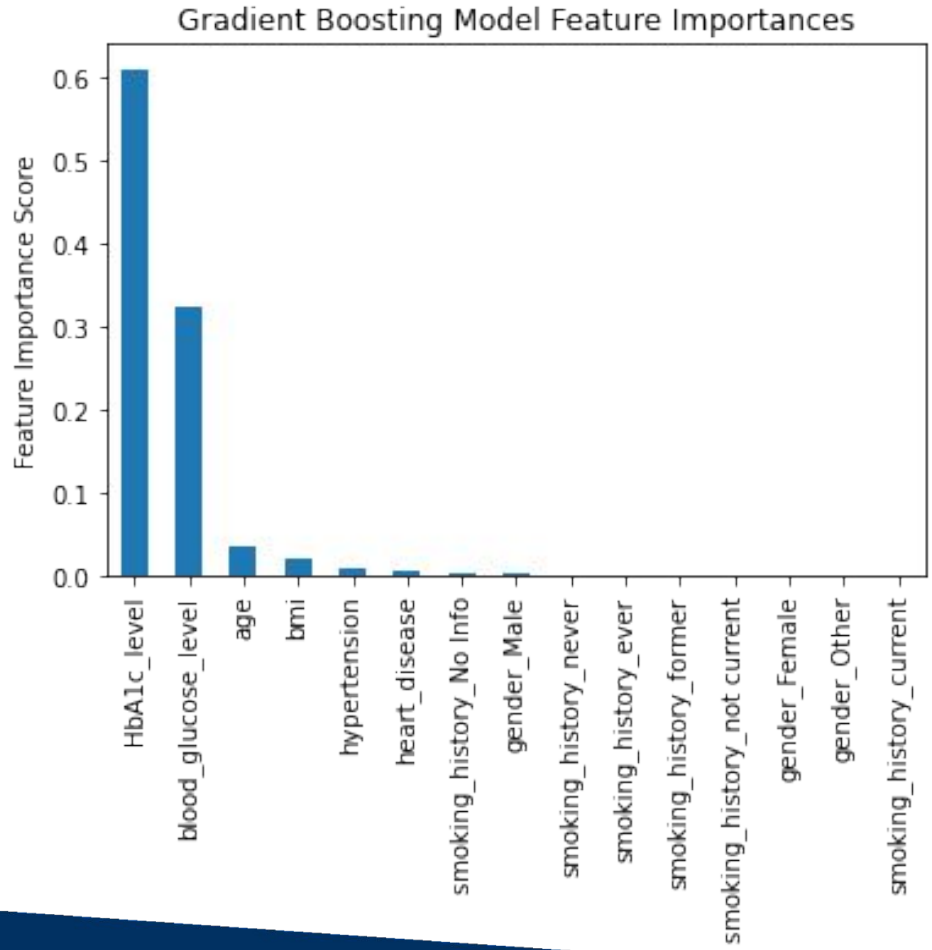


Gradient Boosting Model

Training Dataset Evaluation

- 1647 diabetes observations were predicted as negative by the model
  - Max A1C value of those observations is 6.6
- 5124 diabetes observations in training dataset where A1C <= 6.6
  - 32% of observations were predicted incorrectly

Gradient Boosting Model



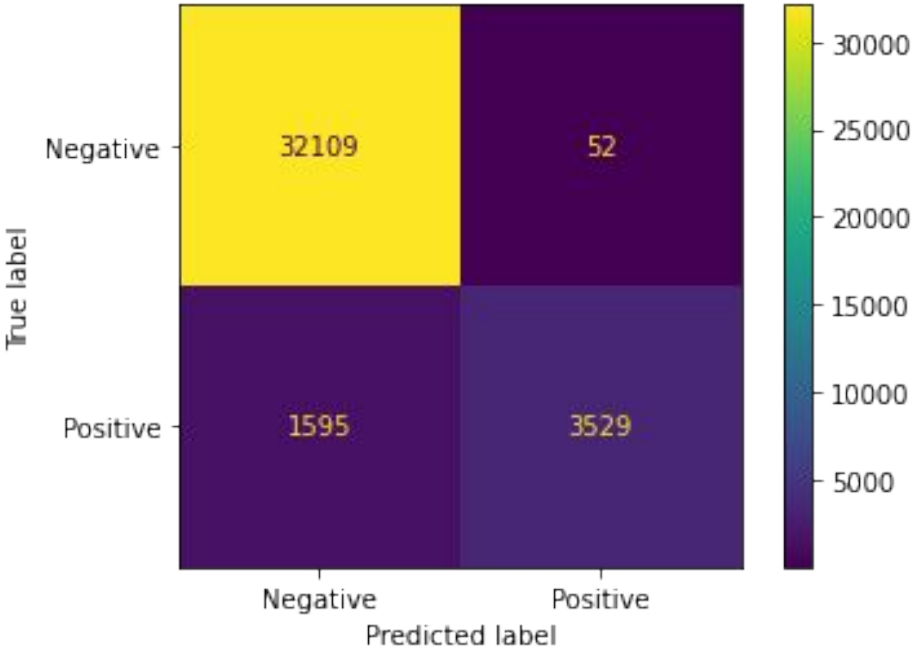
Gradient Boosting Model

Test Dataset Evaluation

	precision	recall	f1-score	support
0	0.95	1.00	0.97	32161
1	0.99	0.69	0.81	5124
accuracy			0.96	37285
macro avg	0.97	0.84	0.89	37285
weighted avg	0.96	0.96	0.95	37285

Gradient Boosting Model

Test Dataset Evaluation – Confusion Matrix



Conclusion

- Our gradient boosting model can predict someone has diabetes with low false positives.
- False negatives are prevalent in our model, specifically with A1C levels lower than 6.6.
- This model can help diagnose people with an initial A1C blood test that they have diabetes before the second A1C blood test.
  - Caveat: A negative diabetes diagnosis by the model does not guarantee one does not have diabetes
- More data and research is needed to see what are other indicators of diabetes, and how the model can be trained to better reduce false negatives.

Questions?

Appendix

Data Source & GitHub (Code Location)

- Data Source: https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset
- Team GitHub page: https://github.com/davidsherman96/w207_final_project_meng_sherman

NeurIPS Checklist

1A. The main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?	**Yes** - details in the dataset description match the data we analyzed
1B. Read the ethics review guidelines and ensured that our paper conforms to them?	**Yes** - the dataset does not raise any concerns about security, discrimination, humans right, among other things
1C. Discussed any potential negative societal impacts of our work?	**Yes** - potential to identify someone as not having diabetes when they actually do have it
1D. Discussed the limitations of our work?	**Yes** - <10% of our sample has diabetes

2A. State full set of assumptions of all theoretical results?	**Yes** - assumptions stated previously and in code
2B. Include complete proofs of all theoretical results?	N/A

3A. Include the code, data, and instructions needed to reproduce the main experimental results?	**Yes** - code's provided on GitHub
3B. Specify all the training details?	**Yes** - details about training/test/validation splits were provided
3C. Report error bars?	N/A
3D. Include amount of compute and type of resources used?	N/A

NeurIPS Checklist Cont.

4A. Cite the creators' existing assets?	**Yes** - source of data was cited
4B. Mention license of the assets?	**N/A** - no mention of copyright or terms of service within data source
4C. Include any new assets either in the supplemental material or as a URL?	**Yes** - data comes from electronic health records from healthcare providers
4D. Discuss whether and how consent was obtained from people whose data you're using?	**Yes** - data gathered (and de-identified) via surveys, medical records, and lab tests
4E. Discuss whether data contains PII or offensive content?	**Yes** - our data does not possess either

5A. Include the full text of instructions given to participants and screenshots?	**N/A**
5B. Describe any potential participant risks, with links to Institutional Review Board, if applicable?	**N/A**
5C. Include the estimated hourly wage paid to participants?	**N/A**

Teammate Contribution Breakdown

Area	Sam	David
Code & Presentation Submission	• Organized and documented all code for our models	• Organized and documented all code for our EDA and built initial draft of deck
Dataset & EDA	• Reviewed David's EDA to begin work on the models	• Described dataset details and key fields and led EDA code/visual efforts
Approach & Models	• Developed functions based off filtered dataset to run our models of interest	• Reviewed Sam's model code and built initial results visuals
Tuning & Improvements	• Conducted hyperparameter tuning and ensemble approaches to attempt to improve results	• Reviewed Sam's tuning work to ensure it made sense
Conclusion & Checklist	• Reviewed David's NeurIPS checklist to ensure our code/project adhered to standard guidelines	• Went through our code and dataset to ensure our work adhered to NeurIPS guidelines